Migration Shocks and Rent Growth in Zoning-Restrictive Housing Markets

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1 Project Motivation and Research Question

The expansion of remote work during and after the COVID-19 period reshaped migration patterns across U.S. regions. However, we do not treat COVID itself as the treatment. Instead, we use pandemic-era variation in migration flows, plausibly induced by remote-work flexibility and changing location preferences, to study the effect of population inflows on local rents.

This project focuses on two main questions:

1. To what extent do population inflows increase local rent growth?
2. Are these rent effects larger in zoning-restrictive housing markets?

The main treatment is population inflow. Zoning restrictiveness is used as a heterogeneity dimension: we test whether the rent effect of population inflows is stronger in areas where housing supply is harder to expand.

2 Data

The unit of analysis is a county-year panel for U.S. counties. The exact time window will be finalized after examining data availability, but the analysis will include both pre-shock and post-shock years.

2.1 Main Variables and Data Sources

All datasets are publicly available, including Zillow ZORI, IRS migration data, ACS, Census Building Permits Survey, WHID, and zoning or land-use regulation datasets.

Role	Dataset	Unit	Variable
Outcome	Zillow ZORI	county-month → county-year	rent growth
Treatment	IRS SOI migration	county-year	migration exposure bin (low / medium / high inflow)
Controls	ACS 5-year	county-year	income, unemployment, renter share, vacancy, housing units
Zoning / Supply Restriction	Census BPS + ACS + zoning datasets	county-year	permits per capita, vacancy rate, zoning restrictiveness

Potential zoning measures include WRLURI or related land-use regulation datasets.

2.2 Outcome Variable

The main outcome variable is local rent growth, measured as the annual change in log rental prices:

$$\Delta \log(Rent_{ct}) = \log(Rent_{ct}) - \log(Rent_{c,t-1}).$$

The primary rent measure will come from the Zillow Observed Rent Index (ZORI). As a robustness check, we may also use ACS median gross rent.

2.3 Treatment Variable

The treatment variable is population inflow, measured using IRS county-to-county migration data.

Rather than treating inflow as a purely continuous variable, we will group counties into migration-exposure bins based on cumulative population inflow during a selected time window. Counties may be classified as low-, medium-, or high-inflow counties.

This bin-based approach allows us to compare rent changes across counties with different levels of migration exposure before and after the migration shock period.

2.4 Heterogeneity Variable

The main heterogeneity variable is zoning restrictiveness. We use zoning and land-use regulation measures to examine whether the effect of population inflows on rents is larger in places where new housing supply is more restricted.

Zoning is not treated as the main treatment. Instead, it is used to study heterogeneous treatment effects.

3 Identification Strategy

Our empirical strategy is a Difference-in-Differences design. We compare counties with different levels of migration inflow before and after the migration shock period.

The baseline specification is:

$$\Delta \log(Rent_{ct}) = \sum_k \beta_k \left(MigrationBin_c^k \times Post_t \right) + X_{ct}\gamma + \alpha_c + \delta_t + \epsilon_{ct},$$

where $MigrationBin_c^k$ indicates whether county c belongs to migration-inflow bin k, $Post_t$ indicates the post-shock period, X_{ct} is a vector of time-varying controls, α_c are county fixed effects, and δ_t are year fixed effects.

The coefficients β_k measure how rent growth changed in counties with different levels of migration inflow relative to low-inflow counties after the migration shock period.

4 Zoning Heterogeneity

To study whether zoning restrictions amplify the effect of migration inflows, we estimate an interaction model:

$$\begin{aligned}\Delta \log(Rent_{ct}) = &\sum_k \beta_k \left(MigrationBin_c^k \times Post_t \right) \\ &+ \sum_k \theta_k \left(MigrationBin_c^k \times Post_t \times ZoningRestrictive_c \right) \\ &+ X_{ct}\gamma + \alpha_c + \delta_t + \epsilon_{ct}.\end{aligned}$$

The key coefficients are θ_k. A positive θ_k would suggest that population inflows generate larger rent increases in zoning-restrictive housing markets.

5 Robustness Checks and Extensions

As an extension, we may use doubly robust or double machine learning methods to flexibly model the relationship between migration inflows, zoning restrictiveness, and rent growth. Machine learning methods such as random forests or gradient boosting can be used for nuisance estimation and compared with the baseline Difference-in-Differences results.

We also plan to conduct several robustness checks, including:

- using alternative rent measures, such as ZORI and ACS median gross rent;
- using alternative migration measures;

- testing different migration-exposure bin definitions;
- conducting pre-trend tests;
- comparing county-level and metro-level specifications.